

Systems & Current Technical Stack

Current architecture and system-level technology mapping across the six live Valence Ops automations.

This document explains the current Valence Ops automation systems in simple terms and maps the technology currently planned or used for each system. The uploaded system context documents are treated as the source of truth. Where a system does not yet have a dedicated context document, the description is based only on the current project definition.

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01

Current Core Stack

Plivo TELEPHONY Inbound/outbound voice infrastructure for the Voice Agent	Sarvam AI AI / LANGUAGE / VOICE AI conversation and Indian-language capabilities
n8n WORKFLOW ORCHESTRATION Runs triggers, business rules, integrations and automation	Supabase DATABASE & INTELLIGENCE Stores operational data, workflow state, events and system intelligence
Sentry MONITORING Tracks application/workflow errors and helps with debugging	Hostinger HOSTING / INFRASTRUCTURE Hosts n8n and other required infrastructure
Meta WhatsApp Cloud API WHATSAPP MESSAGING Customer and internal WhatsApp communication	

SUPERSEDES

Older strategy material mentions Exotel for telephony. The current project decision is Plivo, so this document uses Plivo as the current telephony provider.

02

Voice Agent — AI Voice Receptionist

Voice

Inbound

No context doc yet

WHAT THE SYSTEM DOES

The Voice Agent acts as the clinic's AI receptionist. It handles inbound phone calls and answers questions related to the clinic, its services and other information that a receptionist would normally handle. The goal is to provide an immediate response without requiring a receptionist to answer every call.

HOW IT WORKS

- 01
- Patient/customer calls the clinic number.
- 02
- Plivo handles the telephony connection.
- 03
- The AI conversation layer handles the caller's questions and conversation.
- 04
- The system can use configured clinic information to answer relevant questions.
- 05
- n8n can orchestrate supporting workflows and integrations where required.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Plivo	Inbound telephony and call handling
Sarvam AI	AI conversation, language and voice capabilities
n8n	Workflow orchestration and integrations
Supabase	Database/configuration where clinic or operational data needs to be stored
Sentry	Error monitoring and alerting
Hostinger	Hosting for supporting infrastructure

SOURCE / IMPLEMENTATION NOTE

No dedicated Voice Agent context document has been provided yet. The receptionist behavior above is based on the current project definition.

03 Multi-Channel Intake / Speed-to-Lead — Lead Qualification

WhatsApp

First touch

Schema items to resolve

WHAT THE SYSTEM DOES

This is the first automation a new inbound lead passes through. It engages the lead quickly, identifies the service or treatment they are interested in, asks a small configurable set of qualification questions, scores the lead, records consent and routes the lead to a human representative.

HOW IT WORKS

- 01 Lead arrives through Click-to-WhatsApp sources such as Instagram, Facebook, Google, website or QR.
- 02 WhatsApp webhook receives the inbound message.
- 03 The system checks opt-out status and whether the person is already an existing lead.
- 04 A conversational intake collects service interest and up to three qualification questions.
- 05 Answers are scored and the lead is classified as Hot, Warm or Cold using client-configured rules.
- 06 The qualified information is written to the CRM/database and the lead is assigned to a representative.
- 07 The representative receives a handoff summary.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Meta WhatsApp Cloud API	Inbound and outbound WhatsApp communication
n8n	Webhook handling, conversation flow, qualification logic, routing and automation
Supabase	Lead records, intake state, events, configuration and consent data
Sarvam AI	AI/language layer where conversational interpretation is required
Sentry	Error monitoring and alerting
Hostinger	Hosting for supporting infrastructure

SOURCE / IMPLEMENTATION NOTE

The context document identifies this as MCI and contains several schema/configuration items that must be resolved before build.

04

Quote / Proposal Follow-Up Enforcement

WhatsApp

Sequenced

Email out of scope

WHAT THE SYSTEM DOES

This system makes sure that every proposal or quote sent to a lead receives structured follow-up. It prevents deals from being lost simply because a salesperson forgot to follow up.

HOW IT WORKS

- 01A lead enters the Proposal Sent stage.
- 02The lead is enrolled into a controlled follow-up sequence.
- 03The system sends multiple contextual WhatsApp touches over the defined cycle.
- 04Any reply stops the automated sequence and can trigger a representative handoff.
- 05Negative intent can mark the opportunity as declined.
- 06If there is no response after the full cycle, the sequence is exhausted.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Meta WhatsApp Cloud API	Customer-facing WhatsApp follow-up messages
n8n	Trigger handling, sequence execution, reply handling, SLA monitoring and error handling
Supabase	Leads, sequences, enrollments, touches, events, SLA tracking and logs
Sarvam AI	AI-assisted response handling for questions/free-form conversation where configured
Sentry	Error monitoring and alerting
Hostinger	Hosting for n8n/supporting infrastructure

SOURCE / IMPLEMENTATION NOTE

The current QPF build is WhatsApp-only. Email is explicitly out of scope for the current build.

05

No-Show Recovery System

- WhatsApp
- Appointment-driven
- Hosting reconciled to Hostinger

WHAT THE SYSTEM DOES

The No-Show Recovery System detects when a scheduled appointment is missed and automatically starts a WhatsApp recovery sequence. Its purpose is to recover the missed appointment and route the lead back to a representative when they respond.

HOW IT WORKS

- 01
- Appointment information enters the system.
- 02
- The system sends reminders before the appointment.
- 03
- After the defined grace period, missed appointments are detected.
- 04
- A recovery sequence is started for the no-show.
- 05
- Follow-up touches are sent according to the recovery schedule.
- 06
- A reply stops the automated sequence and can trigger representative handoff.
- 07
- After-hours queued messages can be released when operating hours resume.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Meta WhatsApp Cloud API	Appointment reminders and recovery communication
n8n	Appointment ingestion, reminders, no-show detection, recovery sequences and reply handling
Supabase	Appointments, enrollments, touches, lead events, opt-outs and workflow state
Sentry	Error monitoring and alerting
Hostinger	Current hosting direction for supporting infrastructure

SOURCE / IMPLEMENTATION NOTE

The NSR context document was originally written around n8n Cloud, but the current Valence stack decision is Hostinger for hosting infrastructure.

06 Dead Pipeline Reactivation

- WhatsApp
- Zoho CRM
- Dedicated DB per client
- Testing on hello_world template

WHAT THE SYSTEM DOES

Dead Pipeline Reactivation brings older or inactive leads back into the sales conversation. It is designed for leads that already exist in a client's pipeline but did not convert, while respecting opt-out and marketing-consent requirements.

HOW IT WORKS

- 01 A client's old lead data is imported through a controlled CSV ingestion process.
- 02 Phone and email values are hashed before storage or lookup.
- 03 Leads are deduplicated and checked for opt-out, status and existing enrollments.
- 04 Eligible leads are enrolled into a five-touch reactivation sequence.
- 05 WhatsApp touches are scheduled for T+0, T+2, T+4, T+7 and T+9 days in the current reference design.
- 06 A reply stops the sequence and can trigger representative assignment and handoff.
- 07 The system records events, errors, workflow runs and sequence state.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Meta WhatsApp Cloud API	Outbound reactivation messages
n8n	CSV ingestion, enrollment, scheduled sequence execution, reply detection and handoff
Supabase	Dedicated per-client DPR database, leads, sequences, enrollments, touches, events and logs
Zoho CRM	CRM synchronization and owner updates
Sentry	Current project-wide error monitoring/alerting layer
Hostinger	Current hosting direction for infrastructure

SOURCE / IMPLEMENTATION NOTE

The DPR context specifies a dedicated Supabase project per client for this service. It also specifies Zoho CRM as an integration. In testing mode, outbound cold WhatsApp uses Meta's approved hello_world template.

07 Review Agent

WhatsApp

Post-appointment

Migrations pending

WHAT THE SYSTEM DOES

The Review Agent is the final step after an appointment is completed. It asks the patient to rate their experience. Positive experiences are directed toward a public Google review, while negative experiences are captured privately and routed to clinic staff.

HOW IT WORKS

- 01 The system detects that an appointment has been completed/showed.
- 02 A configurable delay is applied before the review request.
- 03 The patient receives a WhatsApp rating request.
- 04 Excellent/Good responses are directed to a Google review.
- 05 Average/Poor responses are captured as private feedback.
- 06 Clinic staff are notified and a CRM task can be created for negative feedback.

CURRENT TECHNICAL STACK

TECHNOLOGY	USED FOR
Meta WhatsApp Cloud API	Review request, rating interaction and review-link communication
n8n	Trigger, delayed execution, reply handling and routing
Supabase	Review request state, appointment/lead events and feedback data
Zoho CRM	Planned CRM task creation for negative feedback, subject to credential/scope confirmation
Sentry	Error monitoring and alerting
Hostinger	Current hosting direction for infrastructure

SOURCE / IMPLEMENTATION NOTE

The Review Agent context identifies several schema and configuration migrations that must be resolved before production build.

- Coverage Matrix

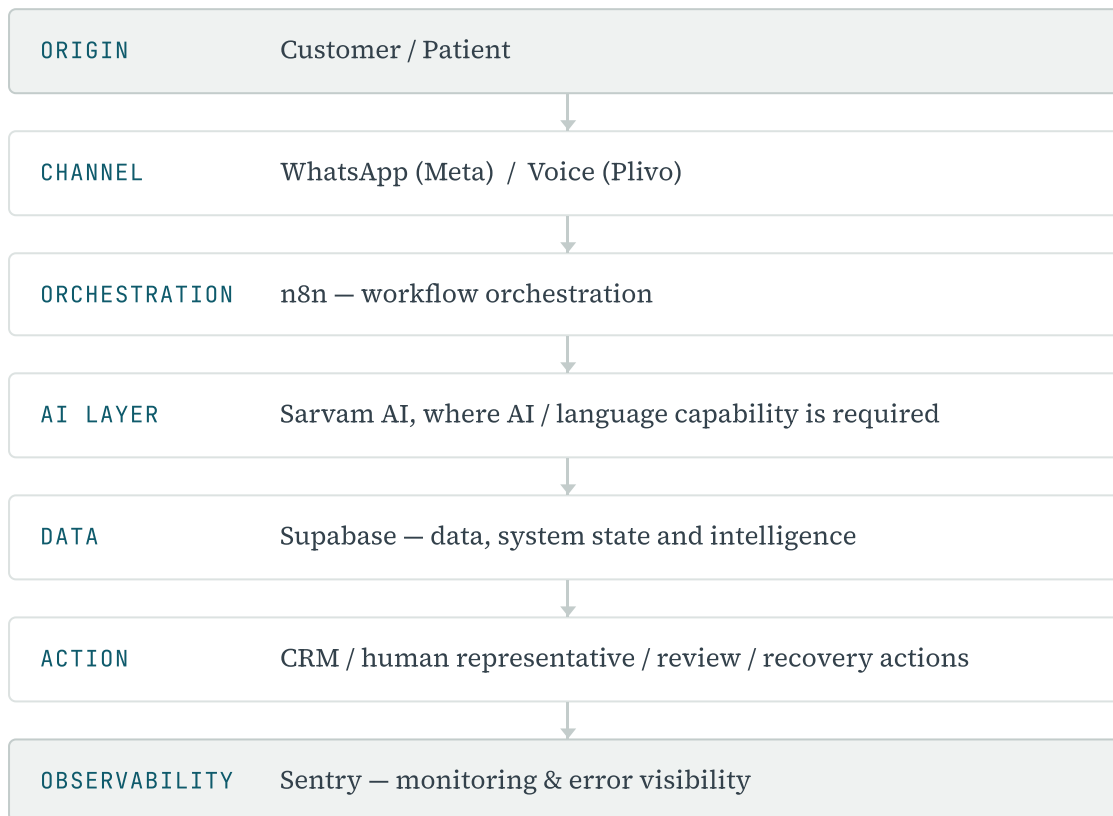
Which technology appears in which system, read directly from the per-system stack tables above.

System	Plivo	WhatsApp Cloud API	n8n	Supabase	Sarvam AI	Zoho CRM	Sentry	Hostinger
Voice Agent	●	○	●	●	●	○	●	●
Multi-Channel Intake	○	●	●	●	●	○	●	●
Quote Follow-Up	○	●	●	●	●	○	●	●
No-Show Recovery	○	●	●	●	○	○	●	●
Pipeline Reactivation	○	●	●	●	○	●	●	●
Review Agent	○	●	●	●	○	●	●	●

● Listed in the system's stack table ○ Not listed for that system

08 Overall Architecture

At a high level, Valence Ops uses communication channels to receive or send customer messages, n8n to execute deterministic workflows, AI services where conversation/language interpretation is needed, Supabase to store operational and intelligence data, and Sentry to monitor failures.



09 Important Current-State Notes

- The stack above reflects the current project direction, not every tool mentioned in older strategy documents.
-
- Plivo is the current telephony choice; older Exotel references are not treated as current.
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- Meta WhatsApp Cloud API is the current WhatsApp messaging layer.
-
- n8n is the common orchestration layer across the systems.
-
- Supabase is the common data layer, but Dead Pipeline Reactivation has a specific per-client dedicated-database architecture.
-
- Sarvam AI is the current AI/language layer chosen by the project; the exact use of AI is system-specific and should not be added to deterministic workflow steps unless the system context requires it.
-
- Sentry is the current monitoring choice. Its exact pricing/plan is not specified in these system context documents and is therefore not stated here.
-
- Some systems still have schema/configuration items to resolve before they should be considered production-ready. Those are explicitly called out rather than being hidden.